

Particle in 1D box

Wavefunctions

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin \frac{n\pi x}{a}$$

 $n: 1, 2, 3, \dots$ n quantum numbers a : Length of the box

PROBABILITY

$$P = \int_{\text{all space}} \psi_m^* \psi_n d\tau \left\{ \begin{array}{l} = 1 \text{ for } m=n \\ \text{(normalized)} \\ = 0 \text{ for } m \neq n \\ \text{(orthogonal)} \end{array} \right.$$

- Wavefunctions are **Orthonormal** (normalization + orthogonality)
- Wavefunction is unique

$$d\tau = dx dy dz \quad \text{Volume element}$$

Wavefunction

$$\Psi_n(x) = \sqrt{\frac{2}{a}} \sin \frac{n\pi x}{a}$$

For state $n = 1$

$$\Psi_1(x) = \sqrt{\frac{2}{a}} \sin \frac{\pi x}{a}$$

For $x = 0$

$$\Psi_1(x) = 0$$

For $x = a/4$

$$\begin{aligned}\Psi_1(x) &= \sqrt{\frac{2}{a}} \sin \frac{\pi}{4} \\ &= \sqrt{\frac{2}{a}} \times \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{a}}\end{aligned}$$

$$|\Psi_1(x)|^2 = \frac{1}{a}$$

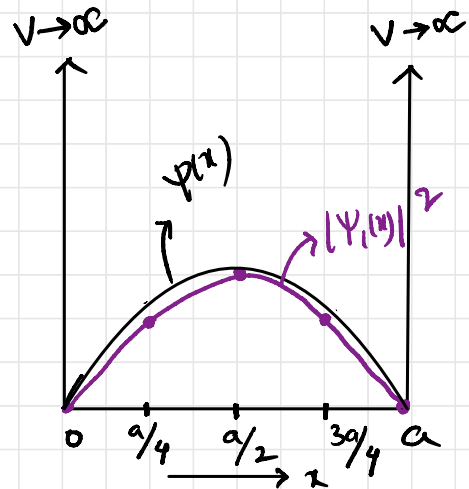
For $x = a/2$

$$\begin{aligned}\Psi_1(x) &= \sqrt{\frac{2}{a}} \sin \frac{\pi}{2} \\ &= \sqrt{\frac{2}{a}}\end{aligned}$$

$$|\Psi_1(x)|^2 = \frac{2}{a}$$

For $x = a$

$$\Psi_1(x) = 0$$



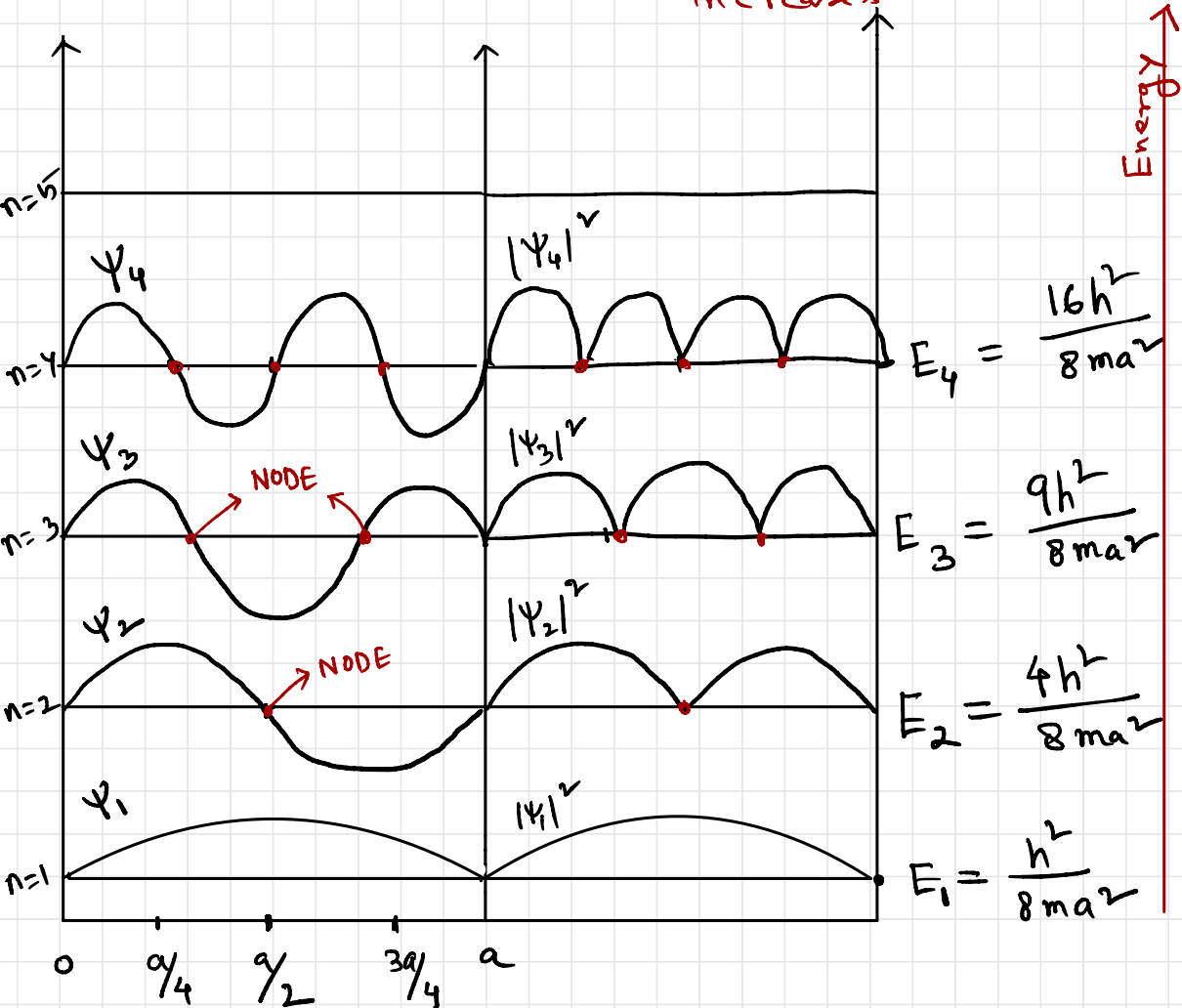
For $x = 3a/4$

$$\begin{aligned}\Psi_1(x) &= \sqrt{\frac{2}{a}} \sin \frac{3\pi}{4} \\ &= \frac{1}{\sqrt{a}}\end{aligned}$$

$$|\Psi_1(x)|^2 = \frac{1}{a}$$

Draw $\psi(x)$ & $|\psi(x)|^2$: $\psi_n(x) = \sqrt{\frac{2}{a}} \sin \frac{n\pi x}{a}$

As quantum numbers increases no. of nodes increases



Wavefunction
 $\psi(x)$

Probability
density
 $|\psi(x)|^2$

Energy
 $E = \frac{n^2 h^2}{8ma^2}$

$E \propto n^2, E \propto \frac{1}{a^2}, E \propto \frac{1}{m}$